Week 2: Daily Journals

Monday:

We began the week with a team meeting to outline our objectives. Following the meeting, Jonah and I investigated a connectivity issue between the Raspberry Pi 4 and the Ubuntu host laptop. The devices interface via Ethernet-over-USB, but the laptop was failing to detect the connection. After systematic troubleshooting, I isolated the root cause to a faulty USB-A-to-USB-C adapter. Replacing the adapter restored physical connectivity. I then attempted to establish an SSH session from the laptop to the Pi, but the connection was unsuccessful. Unable to resolve the SSH issue by the end of the session, I pivoted to developing flight software for the remainder of the day.

Tuesday:

I initiated the day by re-attempting the SSH connection to the Raspberry Pi, which successfully authenticated. With remote access established, I executed my Pi diagnostic test code to verify FlatSat readiness. All parameters were verified successfully except for the mode transition notification, which failed to transmit to the Ubuntu laptop. I refactored the transmission logic, retested the script, and successfully verified all functionalities. Additionally, I implemented a new feature that outputs a test summary to a dedicated log file. Before concluding for the day, I retrieved the fabricated 25-pin D-sub male connector cable from Jack.

Wednesday:

Jonah integrated the 25 individual wire leads into the GPIO Pi Hat while I continued development on the flight software. Kevin provided the software repository used for Software Defined Radio (SDR) data collection. I reviewed and familiarized myself with this codebase, as these modules will soon be integrated into the primary Mode Manager script. Before leaving for the day, I staged the test equipment required to perform continuity and pinout verification on the 25-pin D-sub connector.

Thursday:

Kevin and I initiated pinout verification on the 25-pin connector. During initial continuity testing, we encountered inconsistent data and suspected a structural short circuit, though a visual inspection of the solder joints indicated correct wiring. While the team investigated the electrical anomaly, I continued working on flight software development.

Friday:

Today, I identified the root cause of Thursday's pinout verification failures. The Issue was that our cable design utilized Pin 25 as the reference ground; however, Pin 25 on this specific interface is unassigned (no-connect), resulting in a floating ground and erratic readings. To fix this, we rerouted the ground reference to Pin 23 for the remainder of the procedure. Following this adjustment, all pins successfully passed continuity testing. The remainder of the day was spent organizing the workstation and continuing development on the flight software.